Power Budget - LiDAR Subsystem (Daughter Board)

Team Number: Team 1 (Group 101)
Project Name: **OpenField**
Team Member Names: **Landon Kuhman (LiDAR subsystem)**
Version: **1.0 (10/19/2026)**

All Major Components	**Component Name**	**Part Number**	**Supply Voltage Range**	**Qty.**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
	ToF LiDAR (D500)	STL-19P	+4.5 - 5.5V	1	290	290	mA
	PIC18F57Q43 Nano	DM164150	+4.4 - 5.25V	1	100	100	mA

+12V Rail (unused)	**Component Name**	**Part Number**	**Supply Voltage Range**	**Qty.**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
	None			0	0	0	mA
							mA
							mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	0	mA

Regulator	Not used	N/A	N/A	0	0	0	mA
					Total Remaining Current Available on +12V Rail	0	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ToF LiDAR (D500)	STL-19P	+4.5 - 5.5V	1	290	290	mA
	PIC18F57Q43 Nano	DM164150	+4.4 - 5.25V	1	100	100	mA
							mA
					Subtotal	390	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	487.5	mA
Regulator	+5V LDO Regulator	LM2940CT-5.0	+6V - 26V	1	1000	1000	mA
	Total Remaining Current Available on +5V Rail					512.5	mA

+3.3V Rail (unused)	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	None			0	0	0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	0	mA
Regulator	Not used	N/A	N/A	0	0	0	mA
	Total Remaining Current Available on 3.3V Rail					0	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Will calculate and decide	as a group					
Power Rails Connected to External Power Source 1	Will calculate and decide	as a group					
					Total Remaining Current Available on External Power Source 1		

External Power Source 2 (u	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	None	N/A	N/A	N/A	0	0	mA
Power Rails Connected to External Power Source 2	None	N/A	N/A	0	0	0	mA
					Total Remaining Current Available on External Power Source 2	0	mA

Calculated Battery Life							
	Component Name	**Part Number**	**Supply Voltage Range**		**Capacity (mAh)**	**Required By Regulators**	
	Will calculate and decide	as a group					
					Battery Life		

Notes:

Power source selection and battery life cover the whole robot (all five subsystems), so they will be calculated and decided as a group.

Curiosity Nano current (100 mA) is a conservative allowance for the PIC18F57Q43 at 64 MHz plus the on-board nEDBG debugger; D500 current (290 mA) is the Waveshare/LDRobot rated working current.